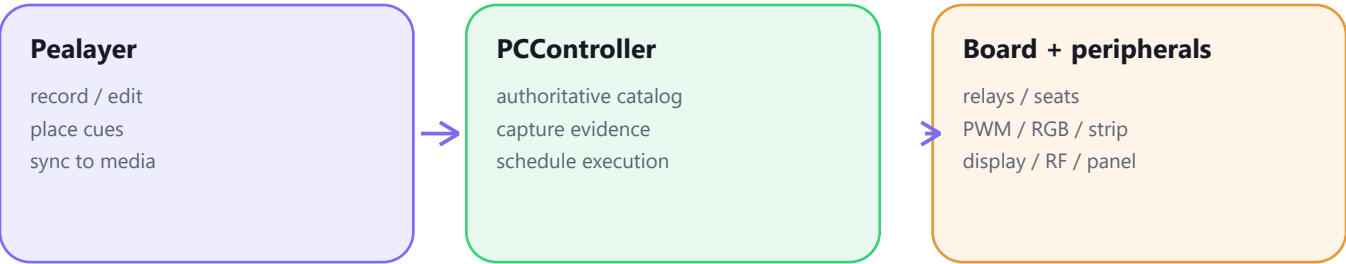


# Effects that move with the movie

A practical guide to recording, editing, placing, and running multi-peripheral effects with Pealayer and PCController



## One durable effect; several temporary execution forms

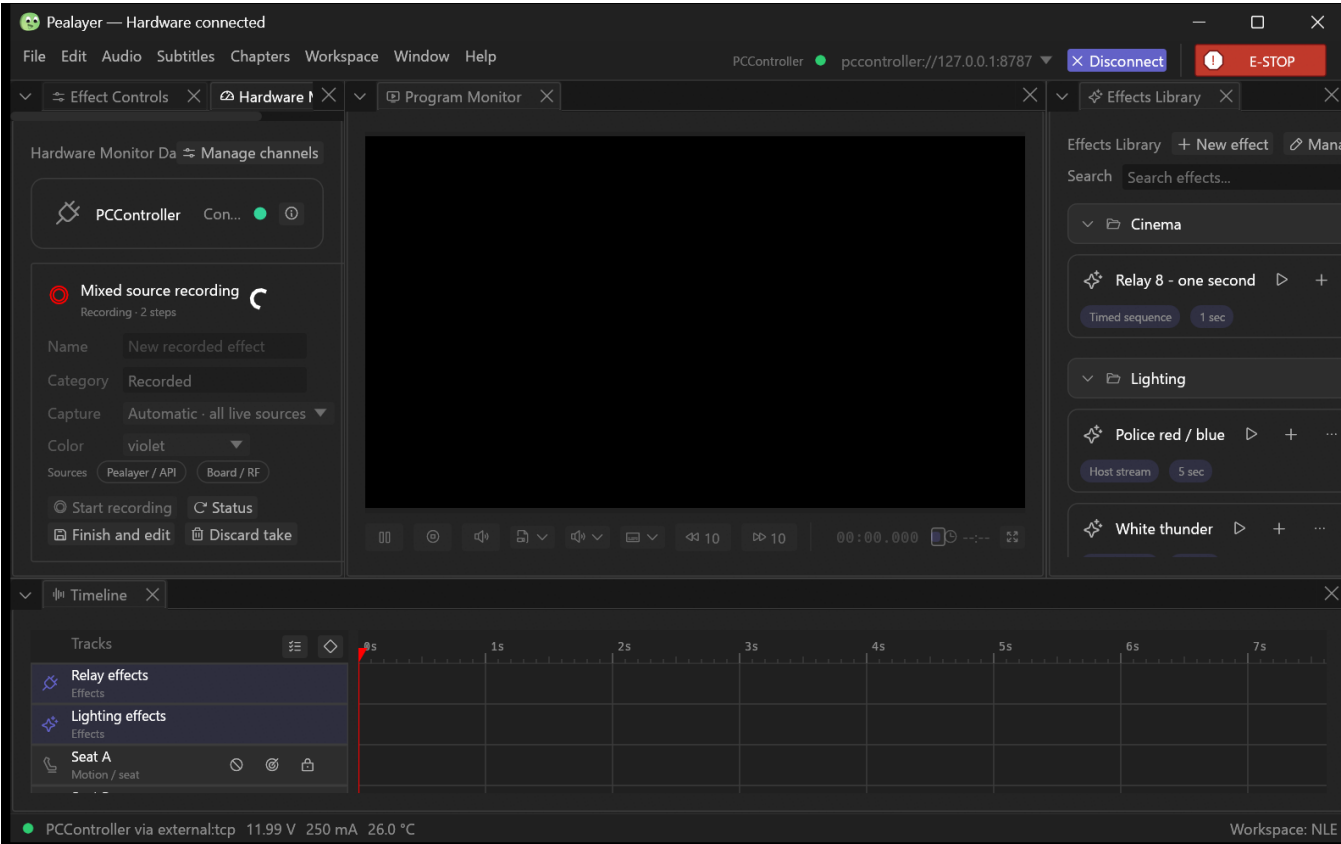
Host RAM, board RAM, and streaming are runtime choices—not separate libraries.  
Every timeline cue stores a stable effect reference plus start, duration, and overrides.

| Verified in this delivery | Result                                                  |
|---------------------------|---------------------------------------------------------|
| Catalog                   | Relay 8 pulse + Police + White thunder + Converging red |
| Mixed-source capture      | Pealayer/API and board-origin relay events in one take  |
| Virtual output proof      | ON:8 followed by OFF:8                                  |
| Timeline                  | 1-second Relay 8 cue placed at 00:00:10.000             |

**Start simple: an effect says what happens; a cue says when it happens in the movie.**

# 1 • Record a real take

Open Hardware Monitor or Effects Library. Choose Record effect, enter a name, and leave capture mode on Automatic · all live sources. Pealayer starts the PCController recorder; it does not create a private second catalog.

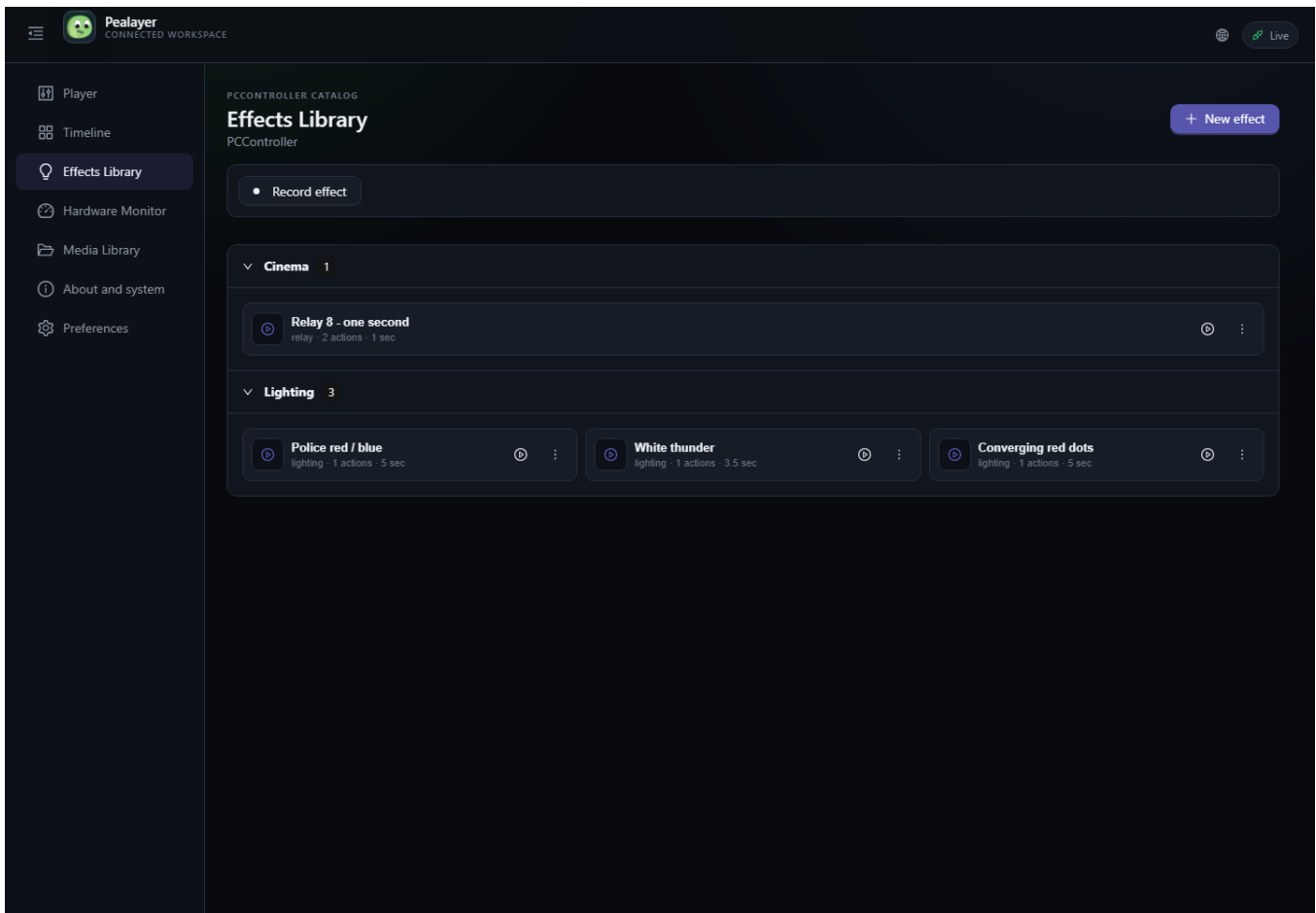


Native egui recorder, capability-driven controls, and the shared PCController catalog.

| Capture mode              | Use it for                                                                                  |
|---------------------------|---------------------------------------------------------------------------------------------|
| Automatic                 | App, Web, API, RF, front-panel, and board events together; best compatible clock is chosen. |
| Device clock              | Strict device acknowledgement timestamps when every action is supported.                    |
| Board-retained relay take | Bounded relay/motion capture kept temporarily in board RAM until Save.                      |

## 2 • Finish, inspect, and edit

Select Finish and edit. The saved definition appears in the PCController-owned Effects Library. Use Manage to edit exact time, channel, value, duration, exit behavior, repeats, and effect presentation. A working draft can remain offline; publishing after reconnection writes through PCController.

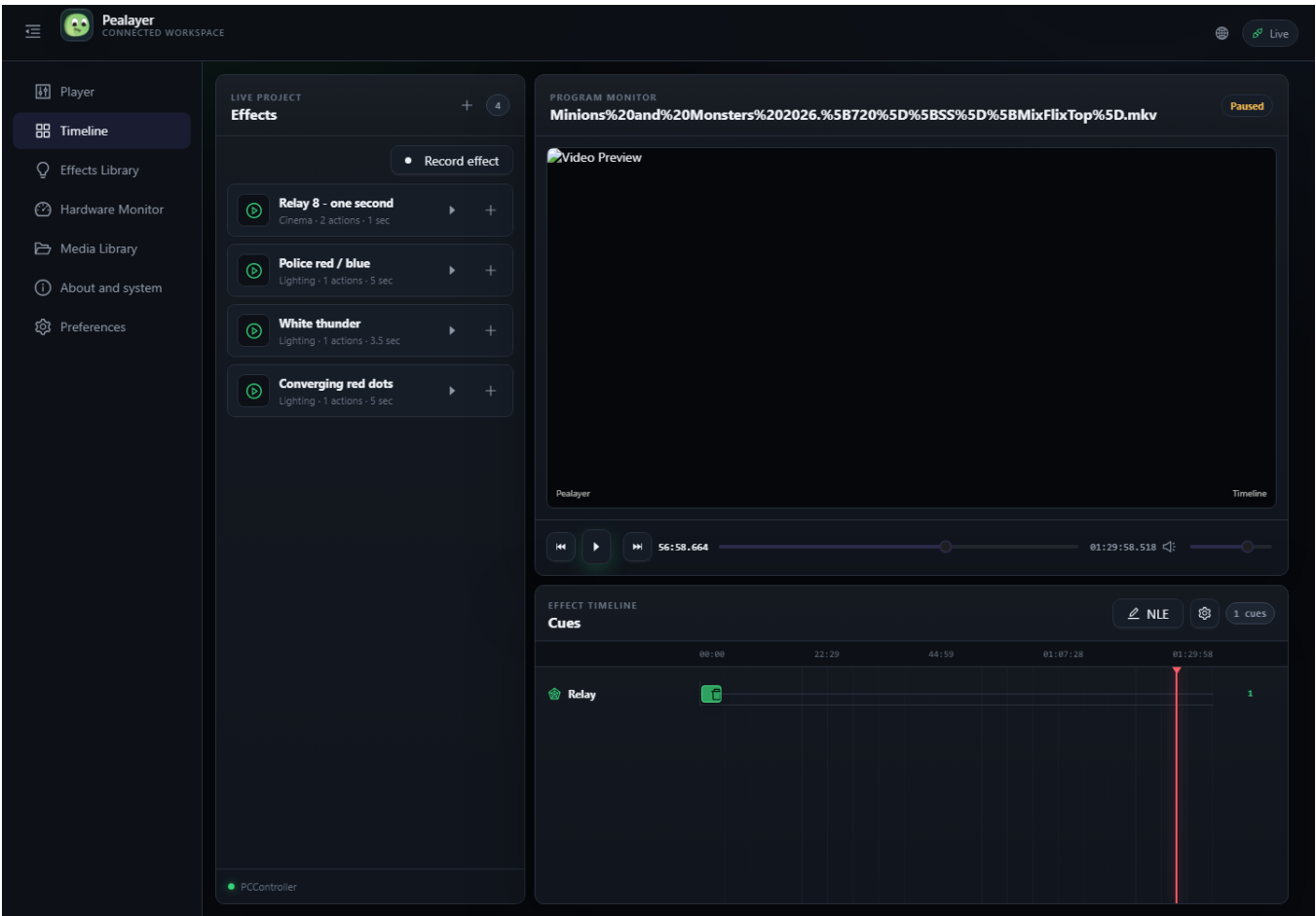


Web Effects Library: the same four live definitions are visible and manageable without hardcoded Pealayer samples.

| Need              | Model                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------|
| One shot          | A single On, Off, Up, Down, Stop, PWM, RF, display, or lighting step.                     |
| Sustain           | Start value plus duration and deterministic exit value.                                   |
| Blink / repeat    | Repeat count and interval; no hand-duplicated steps required.                             |
| Recorded motion   | Edit the captured steps, remove leading delay, quantize, and keep device timing evidence. |
| Mixed peripherals | Several lanes may live in one effect and are compiled into one run plan.                  |

### 3 • Place a cue on the movie

Drag an effect card to its compatible timeline lane. The placed item is a cue. Drag its body to move it; drag either edge to resize a placement. The effect remains reusable and PCController-owned.

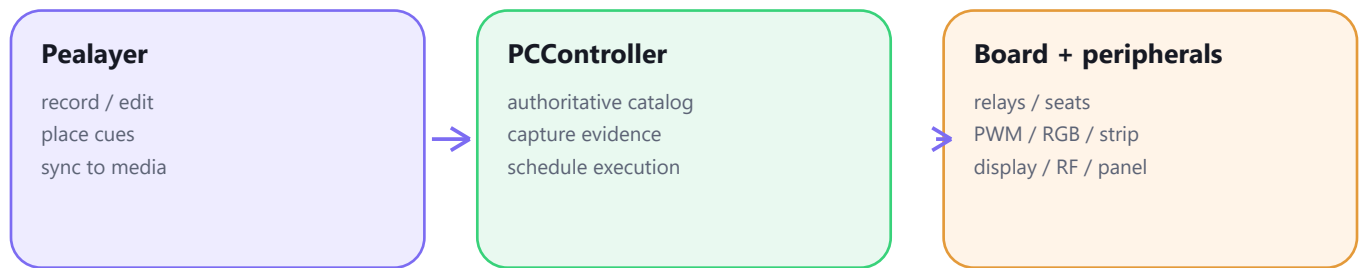


Verified Web timeline with the Relay 8 cue at 00:00:10.000. The thin 1-second block is visible near the lane start because the movie is about 90 minutes long.

#### Worked example

|               |                                           |
|---------------|-------------------------------------------|
| Effect        | Relay 8 — one second                      |
| Step 1        | 0 ms · Relay 8 ON                         |
| Step 2        | 1,000 ms · Relay 8 OFF                    |
| Cue           | Starts at 00:00:10.000; duration 1 second |
| Virtual proof | ON:8 → OFF:8                              |

## 4 · Understand the system



### One durable effect; several temporary execution forms

Host RAM, board RAM, and streaming are runtime choices—not separate libraries.

Every timeline cue stores a stable effect reference plus start, duration, and overrides.

| Term             | Meaning                                                                          |
|------------------|----------------------------------------------------------------------------------|
| Effect           | Reusable, named definition in PCController.                                      |
| Sequence         | Timed steps inside an effect.                                                    |
| Recording / take | Capture session that produces an editable sequence.                              |
| Cue              | One media-timeline placement referring to an effect.                             |
| Step             | One action at an exact offset: relay, motion, PWM, color, RF, display, and more. |
| Lane             | Compatible channel/type grouping used by the editor and timeline.                |
| Run plan         | Temporary capability-aware form staged in host RAM, board RAM, or streamed.      |
| Evidence         | Acknowledgement/event showing an action was actually applied.                    |

**The board and host do not own competing effect libraries. PCController owns the durable catalog; execution may move transparently between host scheduling, device timing, and streamed frames.**

## 5 · Operator checklist and tips

- 01 Confirm Pealayer shows a green PCController connection before recording live hardware.
- 02 Use Automatic capture first; choose a forced clock only while diagnosing timing.
- 03 Record from Pealayer and physical controls at the same time—PCController merges acknowledged evidence.
- 04 Preview with unloaded or supervised motion hardware before using cinema seats.
- 05 Use the three restored lighting examples as editable starting points, not immutable presets.
- 06 Export effects.json before moving a catalog between hosts; importing restores the same stable IDs.
- 07 When hardware is absent, VirtualBoard prints ON:n, OFF:n, and PWMn:value for inspectable proof.
- 08 An E-STOP cancels active effect playback and rejects output commands until released.

### Physical acceptance still matters

The software and VirtualBoard paths are verified. Final cinema-seat and WS2811 acceptance requires the actual controller to be connected, outputs supervised, and the live board profile confirmed. Capture the exact commit, board session, effect ID, timing evidence, and observed output during that test.

**Pealayer coordinates the story. PCController guarantees the hardware action. The effect catalog keeps both interfaces on the same truth.**